LAB SHEET - 4

1. ***Create an XML document containing information about five students, including Student ID, Name, Course, Semester, Email, and Marks.***

Ans 1.

```
<?xml version="1.0" encoding="UTF-8"?>
<students>
   <student>
      <id>101</id>
      <name>Vansh</name>
      <course>BCA</course>
      <semester>3</semester>
      <email>vansh@example.com</email>
      <marks>88</marks>
   </student>
   <student>
      <id>102</id>
      <name>Aman</name>
      <course>BCA</course>
      <semester>3</semester>
      <email>aman@example.com</email>
      <marks>76</marks>
   </student>
   <student>
      <id>103</id>
      <name>Rohan</name>
      <course>BCA</course>
      <semester>3</semester>
```

```
    <email>rohan@example.com</email>
    <marks>92</marks>
  </student>
  <student>
    <id>104</id>
    <name>Priya</name>
    <course>BCA</course>
    <semester>3</semester>
    <email>priya@example.com</email>
    <marks>81</marks>
  </student>
  <student>
    <id>105</id>
    <name>Sahil</name>
    <course>BCA</course>
    <semester>3</semester>
    <email>sahil@example.com</email>
    <marks>69</marks>
  </student>
</students>
```

2. ***Create an XML document to store details of at least five books, including Book ID, Title, Author, Publisher, Price, and Category.***

Ans 2.

```
<?xml version="1.0" encoding="UTF-8"?>
<bookstore>
  <book>
    <id>B101</id>
    <title>HTML and CSS</title>
    <author>Jon Duckett</author>
```

```
    <publisher>Wiley</publisher>
    <price>650</price>
    <category>Web Development</category>
</book>
<book>
    <id>B102</id>
    <title>JavaScript: The Definitive Guide</title>
    <author>David Flanagan</author>
    <publisher>O'Reilly</publisher>
    <price>950</price>
    <category>Programming</category>
</book>
<book>
    <id>B103</id>
    <title>Clean Code</title>
    <author>Robert C. Martin</author>
    <publisher>Prentice Hall</publisher>
    <price>800</price>
    <category>Software Engineering</category>
</book>
<book>
    <id>B104</id>
    <title>Python Crash Course</title>
    <author>Eric Matthes</author>
    <publisher>No Starch Press</publisher>
    <price>720</price>
    <category>Programming</category>
</book>
```

```
  <book>
    <id>B105</id>
    <title>Database System Concepts</title>
    <author>Abraham Silberschatz</author>
    <publisher>McGraw-Hill</publisher>
    <price>890</price>
    <category>Database</category>
  </book>
</bookstore>
```

3. ***Create an HTML webpage containing a list of students and use JavaScript DOM methods such as getElementById(), getElementsByTagName(), and querySelector() to access and display the information.***

Ans 3.

```
<ul id="studentList">
  <li class="std">Sadvansh</li>
  <li class="std">Rahul</li>
  <li class="std">Ankit</li>
</ul>
<button onclick="showStudents()">Display Students</button>
<div id="output"></div>

<script>
  function showStudents() {
    let list = document.getElementById("studentList");
    let items = document.getElementsByTagName("li");
    let firstStudent = document.querySelector(".std").innerText;

    document.getElementById("output").innerHTML =
      "<br>Total Students: " + items.length +
```

```
        "<br>First Student (querySelector): " + firstStudent +
        "<br>Second Student (getElementsByTagName): " + items[1].innerText;
    }
</script>
```

4. ***Create an HTML page containing a heading, paragraph, and button. Use JavaScript DOM to change the heading text, paragraph content, and background color when the button is clicked.***

Ans 4.

```
<h1 id="heading">Original Heading</h1>
<p id="para">Original paragraph text.</p>
<button onclick="changeContent()">Change Details</button>

<script>
    function changeContent() {
        document.getElementById("heading").innerText = "Updated Heading Text";
        document.getElementById("para").innerText = "Paragraph text changed dynamically via DOM.";
        document.body.style.backgroundColor = "#e0f7fa";
    }
</script>
```

5***. Create an XML file containing student records and use JavaScript/DOM concepts to read and display the student information dynamically on an HTML webpage.***

Ans 5.

```
<button onclick="loadXMLData()">Display Students from XML</button>
<table border="1" id="recordTable" style="margin-top:10px;">
    <tr><th>ID</th><th>Name</th><th>Course</th><th>Marks</th></tr>
</table>

<script>
```

```
    function loadXMLData() {
        let xmlString = `
        <students>

<student><id>101</id><name>Sadvansh</name><course>BCA</course><marks
>90</marks></student>

<student><id>102</id><name>Aditya</name><course>BCA</course><marks>85
</marks></student>
        </students>`;

        let parser = new DOMParser();
        let xmlDoc = parser.parseFromString(xmlString, "text/xml");
        let students = xmlDoc.getElementsByTagName("student");
        let table = document.getElementById("recordTable");

        for (let i = 0; i < students.length; i++) {
            let id = students[i].getElementsByTagName("id")
[0].childNodes[0].nodeValue;
            let name = students[i].getElementsByTagName("name")
[0].childNodes[0].nodeValue;
            let course = students[i].getElementsByTagName("course")
[0].childNodes[0].nodeValue;
            let marks = students[i].getElementsByTagName("marks")
[0].childNodes[0].nodeValue;

            let row = `<tr><td>${id}</td><td>${name}</td><td>${course}</td><td>$
{marks}</td></tr>`;
            table.innerHTML += row;
        }
    }
```

</script>